# Resume Structured Schema

**Purpose:** Standard, AI-ready structure of one parsed resume (Candidate profile). Produced by the resume parser; read by the ATS, screening, interview and scoring engines.

**Files:** `data/schemas/resume_schema.json` (JSON Schema draft-07), `data/samples/sample_resume.json`, validated with `utils/schema_validator.py`.

## Design rules

- Required: candidate_id, personal.full_name, personal.email, skills, experience, education.
- Missing data is null or an empty list, never guessed (a fresher has an empty experience list).
- Original and normalized values stored together (name / normalized_name, designation / normalized_designation).
- Dates use YYYY-MM. A current job has end_date null and is_current true.
- No gender, age or photo fields (bias-reduced scoring). AI interview, recording and voice-cloning consent is in the consent block.
- additionalProperties is false on every object, so misspelled fields are rejected.

Legend: Req = required (Y/-). A trailing ? on a type means the value may be null.

## Top-level

| Field | Type | Req | Rules |
|---|---|---|---|
| schema_version | string | Y | must be 1.0 |
| candidate_id | string | Y | not empty |
| personal | object | Y | see personal |
| headline, summary, current_designation | string? | - | |
| total_experience_years | number? | - | 0 to 60 |
| skills / experience / education | arrays | Y | see tables below |
| certifications, projects, languages | arrays | - | see tables below |
| preferences, consent, parsing_meta | objects | - | see tables below |

## personal

| Field | Type | Req | Rules |
|---|---|---|---|
| full_name | string | Y | not empty |
| email | string | Y | email format |
| phone | string? | - | |
| location.city / state / country | string? | - | |
| links.linkedin / github / portfolio | string? | - | |

## skill

| Field | Type | Req | Rules |
|---|---|---|---|
| name | string | Y | as written in resume |
| normalized_name | string? | - | lower-case canonical, e.g. node.js |
| category | string | - | programming_language, framework, database, cloud_devops, tool, design, business_domain, soft_skill, other |
| proficiency | string? | - | beginner, intermediate, advanced, expert |
| years_used | number? | - | min 0 |
| last_used | string? | - | YYYY-MM |
| source | string | - | skills_section, experience, project, certification, inferred |
| confidence | number | - | 0 to 1 |

## experience

| Field | Type | Req | Rules |
|---|---|---|---|
| company, designation | string | Y | not empty; designation as written |
| start_date | string? | Y | YYYY-MM or null |
| end_date | string? | - | YYYY-MM; null if current |
| normalized_designation | string? | - | standard title, e.g. software engineer |
| employment_type | string? | - | full_time, part_time, internship, contract, freelance |
| industry, location, description | string? | - | |
| is_current | boolean | - | |
| duration_months | integer? | - | min 0 |
| responsibilities, achievements, skills_used | array of string | - | |

## education

| Field | Type | Req | Rules |
|---|---|---|---|
| degree, institution | string | Y | not empty |
| level | string | - | secondary, higher_secondary, diploma, bachelor, master, doctorate, other |
| field_of_study | string? | - | |
| start_year, end_year | integer? | - | 1950 to 2100 |
| grade.value | number? | - | |
| grade.scale | string? | - | cgpa_10, cgpa_4, percentage, other |

## certification / project / language

| Field | Type | Req | Rules |
|---|---|---|---|
| certification.name | string | Y | not empty |
| certification.issuer, credential_id, url | string? | - | |
| certification.issue_date, expiry_date | string? | - | YYYY-MM |
| project.name | string | Y | not empty |
| project.description, url | string? | - | |
| project.skills_used | array of string | - | |
| language.language | string | Y | |
| language.proficiency | string? | - | basic, conversational, fluent, native |

## preferences / consent / parsing_meta

| Field | Type | Req | Rules |
|---|---|---|---|
| preferences.notice_period_days | integer? | - | min 0 |
| preferences.expected_salary | object | - | amount (number?, min 0), currency (3-letter), period (monthly, yearly) |
| preferences.preferred_locations | array of string | - | |
| preferences.preferred_call_language | string? | - | |
| preferences.preferred_voice_gender | string? | - | male, female, no_preference |
| consent.ai_interview, recording, voice_cloning | boolean | - | |
| consent.consent_given_at | string? | - | date-time |
| parsing_meta.source_file, parser_version | string | - | |
| parsing_meta.file_type | string | - | pdf, doc, docx |
| parsing_meta.parsed_at | string | - | date-time |
| parsing_meta.overall_confidence | number | - | 0 to 1 |
| parsing_meta.warnings | array of string | - | |